Global Warming and the Elderly: A Socio-Ecological Framework

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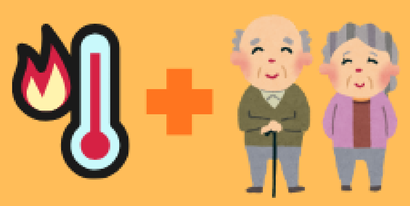
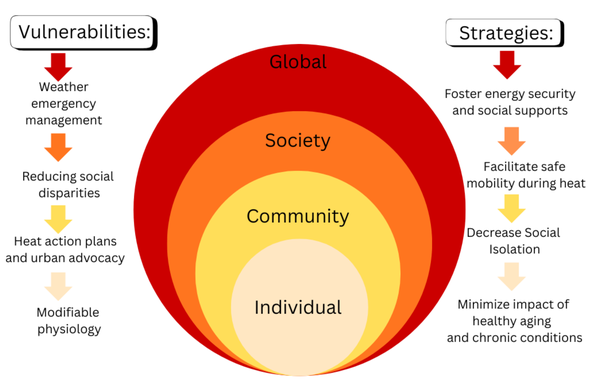
Problem Statement:
Heat-related mortality among individuals aged 65 and over increased

167%

compared to the 1990s.



Hotter temps & Aging population



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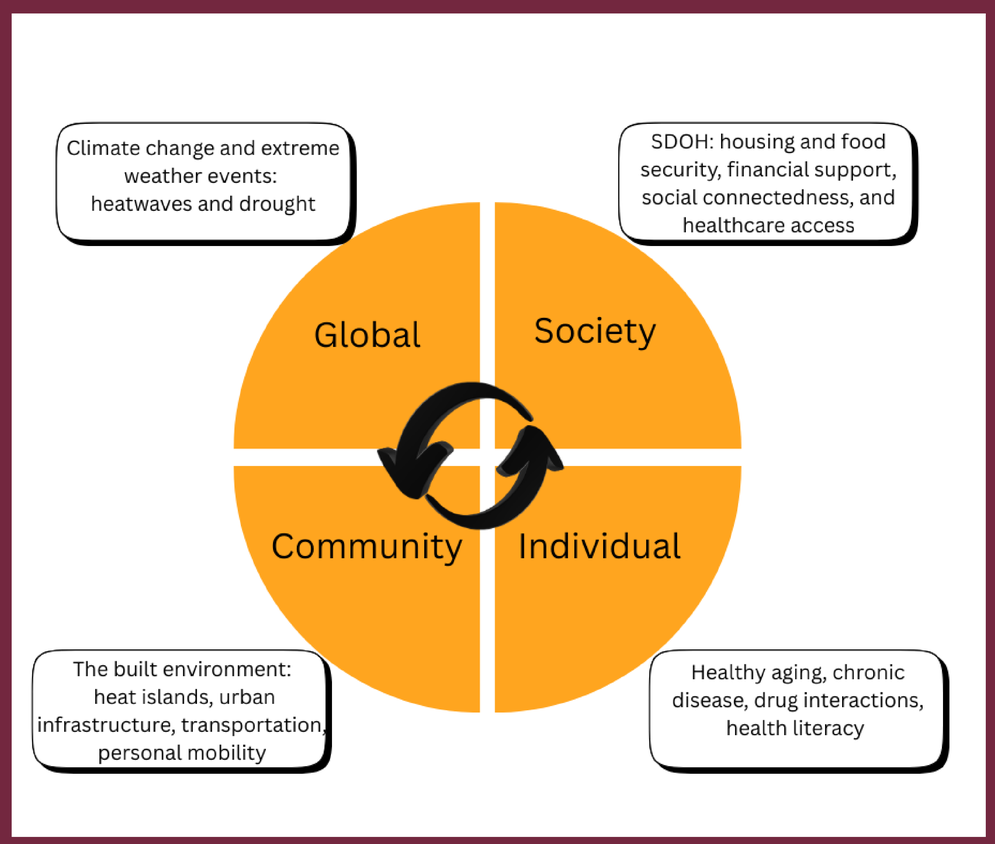
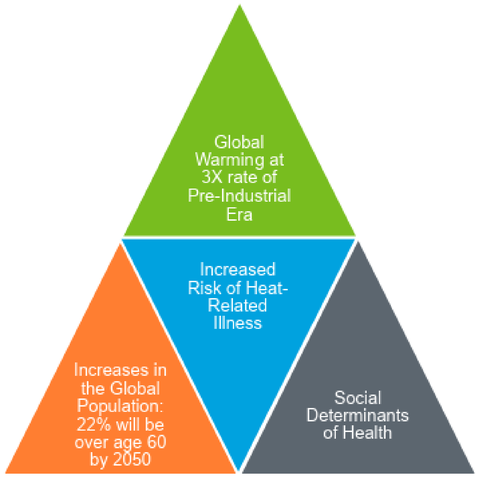
Effective mitigation of heat illness in aging adults includes patient education, community-wide warning systems, social support, and structural redesign for personal mobility and safe, affordable housing.





AGING AND HEAT TOLERANCE:
Physiological changes that occur with healthy aging: include

- Declines in bone density and muscle mass,
- Changes in vision, hearing acuity, and in some cases cognitive slowing.
- Maximum heart rate, cardiac output, and oxygen uptake decrease.
- Bodies are less effective at thermoregulation.
- Delayed threshold for sweating, and total body water decreases.

TWO TYPES OF HEAT STROKE:
Classical: Hot, dry skin with a rectal temperature over 104 degrees F, leading to confusion, loss of consciousness, and convulsions.
Exertional: Brought on by intense physical activity in extreme heat, characterized by sweating, rhabdomyolysis, rapid heartrate, and elevated core temperatures over 40 degrees F.
Classical heatstroke is the more dangerous condition due to its insidious onset.

CONTACT BURNS
In extremely hot climates, falls on paved surfaces can lead to contact burns.

EFFECTS OF MEDICATIONS ON HEAT TOLERANCE:
Medications which may reduce tolerance to extreme temperatures include **opiates, benzodiazepines, antidepressants (SSRIs, TCAs, antipsychotics, and anticonvulsants, diuretics, calcium channel blockers, and anticholinergics.**
OTC medications (NSAIDs) and alcohol also impair the body's ability to tolerate heat.

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